

JESS Thermodynamic Database v8.9

Chemical species in reactions with Tungsten

24-Jan-23

Charge	CAS	Count	Molecular formula	Mol. mass
2OH2MePropan-1				
-1	---	197	C(4)H(7)O(3)	103.098
2OHButan-1 2-Hydroxybutanoate ion				
-1	---	36	C(4)H(7)O(3)	103.098
35DiNitrCat-2				
-2	---	4	C(6)H(2)N(2)O(6)	198.092
3CaO.WO3 (s) Calcium orthotungstate; Tricalcium tungsten hexaoxide				
0	---	1	Ca(3)O(6)W(1)	400.070
Ag+1 Silver(I) ion				
1	14701-21-4	1534	Ag(1)	107.870
Ag+1 (2) _WO4-2_ (s) Silver tungstate; Silver(VI) tungstate				
0	13465-93-5	3	Ag(2)O(4)W(1)	463.578
Ag (s) Silver; Silver, cubic				
0	7440-22-4	182	Ag(1)	107.870
Asp-2 Aspartate ion; Aminosuccinate ion; Asparaglate ion; 2-Aminobutanedioate ion; L-2-Aminobutanedioate ion				
-2	63-71-8	776	C(4)H(5)N(1)O(4)	131.088
Ba+2 _WO4-2_ (s)				
0	---	1	Ba(1)O(4)W(1)	385.168

Ba (s) Barium; Barium, cubic; Barium, lambda trans.				
0	7440-39-3	64	Ba (1)	137.330
Be+2_WO4-2_ (s) Beryllium tungstate				
0	---	1	Be (1) O (4) W (1)	256.850
Be (s) Beryllium; Beryllium, hexagonal; Beryllium, alpha; Beryllium, hcp				
0	7440-41-7	44	Be (1)	9.01220
Br2 (l) Bromine, liquid				
0	---	285	Br (2)	159.808
C (s) Carbon; Graphite; Carbon, hexagonal				
0	7440-44-0	530	C (1)	12.0110
Ca+2 Calcium(II) ion				
2	14127-61-8	2161	Ca (1)	40.0780
Ca+2_WO4-2_ (s) Calcium tungstate				
0	7790-75-2	2	Ca (1) O (4) W (1)	287.916
Ca (s) Calcium; Calcium, cubic; Calcium, fcc; Calcium, alpha				
0	7440-70-2	184	Ca (1)	40.0780
Cd+2_WO4-2_ (s) Cadmium tungstate				
0	---	1	Cd (1) O (4) W (1)	360.248
Cd (s) Cadmium; Cadmium, hexagonal; Cadmium, gamma				
0	7440-43-9	97	Cd (1)	112.410

Cl-1 Chloride ion				
-1	16887-00-6	2242	Cl (1)	35.4530
Cl2 (g) Chlorine gas				
0	7782-50-5	589	Cl (2)	70.9060
Co+2_WO4-2_ (s) Cobalt tungstate				
0	---	1	Co (1) O (4) W (1)	306.771
Co (s) Cobalt; Cobalt, hcp; Cobalt, alpha				
0	7440-48-4	133	Co (1)	58.9330
Cu+2_WO4-2_ (s) Copper tungstate				
0	---	1	Cu (1) O (4) W (1)	311.384
Cu (s) Copper; Copper, cubic				
0	7440-50-8	254	Cu (1)	63.5460
Cys-2 Cysteinate ion; L-Cysteinate ion; beta-mercaptoalanate ion; 2-amino-3-mercaptopropanoate ion; 2-amino-3-mercaptopropionate ion; alpha-amino-beta-thiolpropionate ion				
-2	104170-20-9	673	C (3) H (5) N (1) O (2) S (1)	119.138
DLyxose D-Lyxose; alpha-D-Lyxose; Lyxose D				
0	1114-34-7	7	C (5) H (10) O (5)	150.131
DMannose D-Mannose; Seminose; Carubinoase				
0	3458-28-4	16	C (6) H (12) O (6)	180.158
e-1 Electron				
-1	---	3399	E (1)	0.000000

EDDA-2 Ethylenediiminodiacetate ion; Ethylenediamine-N,N'-diacetate ion; N,N'-ethylene-diglycinate ion; EDDA ion; N,N'-1,2-ethanediylbisglycinate ion; N,N'-Ethylenediaminediacetate anion				
-2	---	154	C(6)H(10)N(2)O(4)	174.156
EDDA-2_WO4-2				
-4	---	1	C(6)H(10)N(2)O(8)W(1)	421.994
EDTA-4 Ethylenediaminetetraacetate anion				
-4	---	1457	C(10)H(12)N(2)O(8)	288.214
F2 (g) Fluorine gas				
0	7782-41-4	259	F(2)	37.9968
Fe+2 Iron(II) ion; Ferrous ion				
2	15438-31-0	1769	Fe(1)	55.8452
Fe+2_WO4-2_(s) Iron tungstate				
0	---	2	Fe(1)O(4)W(1)	303.683
Fe(s) Iron; Iron, cubic; Iron, bcc				
0	7439-89-6	184	Fe(1)	55.8452
Glu-2 Glutamate ion; 2-aminopentanedioate ion; L-Glutamate ion; alpha-aminoglutarate ion; 1-aminopropane-1,3-dicarboxylate ion				
-2	138-18-1	650	C(5)H(7)N(1)O(4)	145.115
Glycolic-1 Glycolate ion				
-1	57122-18-6	342	C(2)H(3)O(3)	75.0440
H+1 Hydrogen ion; Proton				
1	12408-02-5	31679	H(1)	1.00794

H+1_BenzHydroxam-1 Benzohydroxamic acid				
0	495-18-1	3	C(7)H(7)N(1)O(2)	137.138
H+1_W6O21-6				
-5	---	2	H(1)O(21)W(6)	1440.04
H+1_WO3_EDTA-4				
-3	---	1	C(10)H(13)N(2)O(11)W(1)	521.060
H+1_WO4-2				
-1	---	2	H(1)O(4)W(1)	248.846
H+1(2)_Cat-2 Pyrocatechol; Catechol; 1,2-Benzenediol; 1,2-Dihydroxybenzene; CAT				
0	120-80-9	49	C(6)H(6)O(2)	110.112
H+1(2)_DLyxose_WO4-2(2)				
-2	---	1	C(5)H(12)O(13)W(2)	647.823
H+1(2)_DMannose_WO4-2(2)				
-2	---	1	C(6)H(14)O(14)W(2)	677.849
H+1(2)_IDA-2_WO4-2				
-2	---	1	C(4)H(7)N(1)O(8)W(1)	380.942
H+1(2)_LRhamnose_WO4-2(2)				
-2	---	1	C(6)H(14)O(13)W(2)	661.849
H+1(2)_MIDA-2_WO4-2				
-2	---	1	C(5)H(9)N(1)O(8)W(1)	394.968
H+1(2)_Oxalic-2				
0	144-62-7	33	C(2)H(2)O(4)	90.0355
H+1(2)_WO3_Lactic-1				
1	---	1	C(3)H(7)O(6)W(1)	322.925

H+1 (2) _WO3_Lactic-1 (2)				
0	---	1	C (6) H (12) O (9) W (1)	411.996
H+1 (2) _WO3_Mandelic-1				
1	---	1	C (8) H (9) O (6) W (1)	384.996
H+1 (2) _WO3_Mandelic-1 (2)				
0	---	1	C (16) H (16) O (9) W (1)	536.138
H+1 (2) _WO3_Oxalic-2				
0	---	2	C (2) H (2) O (7) W (1)	321.874
H+1 (2) _WO4-2 Hydrogen wolframate; Tungstic acid				
0	---	3	H (2) O (4) W (1)	249.854
H+1 (2) _WO4-2_(s)				
0	---	2	H (2) O (4) W (1)	249.854
H+1 (2) _WO4-2_EDTA-4				
-4	---	2	C (10) H (14) N (2) O (12) W (1)	538.067
H+1 (2) _WO4-2_NTA-3				
-3	---	1	C (6) H (8) N (1) O (10) W (1)	437.970
H+1 (3) _LRhamnose_WO4-2 (2)				
-1	---	1	C (6) H (15) O (13) W (2)	662.857
H+1 (3) _WO3_Lactic-1				
2	---	1	C (3) H (8) O (6) W (1)	323.933
H+1 (3) _WO3_Lactic-1 (2)				
1	---	1	C (6) H (13) O (9) W (1)	413.004
H+1 (3) _WO3_Mandelic-1				
2	---	1	C (8) H (10) O (6) W (1)	386.004

H+1 (3) _WO3_Mandelic-1 (2)				
1	---	1	C (16) H (17) O (9) W (1)	537.146
H+1 (3) _WO3 (2) _Lactic-1 (2)				
1	---	1	C (6) H (13) O (12) W (2)	644.842
H+1 (4) _WO4-2 (2) _EDTA-4				
-4	---	2	C (10) H (16) N (2) O (16) W (2)	787.921
H2 (g) Hydrogen gas; Hydrogen				
0	1333-74-0	1564	H (2)	2.01588
H2O Water				
0	7732-18-5	5947	H (2) O (1)	18.0153
Hg2+2 Mercurous ion				
2	12596-26-8	78	Hg (2)	401.184
Hg2+2 _WO4-2 (s) Mercury(I) tungstate				
0	---	1	Hg (2) O (4) W (1)	649.022
IDA-2 Iminodiacetate ion				
-2	28528-43-0	428	C (4) H (5) N (1) O (4)	131.088
IDA-2 _WO4-2				
-4	---	1	C (4) H (5) N (1) O (8) W (1)	378.926
Lactic-1 Lactate ion				
-1	---	643	C (3) H (5) O (3)	89.0709
LRhamnose L-Rhamnose; L-Mannomethylose; Isodulcitol; alpha-6-Deoxy-L(+)-mannose; 6-Deoxymannose				
0	3615-41-6	8	C (6) H (12) O (5)	164.158

Malonic-2 Malonate ion				
-2	156-80-9	417	$C(3)H(2)O(4)$	102.047
Mandelic-1 α -Hydroxybenzeneacetate ion; Mandelate ion; L-Phenylhydroxyacetate ion; Hydroxybenzene acetate ion; Phenylglycolate ion; Uromalinate ion; Amaygdalate ion				
-1	769-61-9	132	$C(8)H(7)O(3)$	151.142
Mg+2_WO4-2_(s) Magnesium tungstate				
0	13573-11-0	1	$Mg(1)O(4)W(1)$	272.143
Mg(s) Magnesium; Magnesium, hexagonal				
0	7439-95-4	154	$Mg(1)$	24.3050
MIDA-2				
-2	---	147	$C(5)H(7)N(1)O(4)$	145.115
Mn+2 Manganese(II) ion; Manganous ion				
2	16397-91-4	1956	$Mn(1)$	54.9380
Mn+2_WO4-2_(s) Manganese tungstate				
0	---	2	$Mn(1)O(4)W(1)$	302.776
Mn(s) Manganese; Manganese, alpha; Manganese, complex bcc				
0	7439-96-5	72	$Mn(1)$	54.9380
MoO3_NTA-3				
-3	---	2	$C(6)H(6)Mo(1)N(1)O(9)$	332.077
MoO4-2 Molybdate(VI) ion				
-2	14259-85-9	281	$Mo(1)O(4)$	159.960

Na+1 (2)_WO4-2_(s) Sodium tungstate				
0	---	1	Na (2) O (4) W (1)	293.818
Na (s) Sodium; Sodium, cubic				
0	7440-23-5	209	Na (1)	22.9900
Ni+2_WO4-2_(s) Nickel tungstate				
0	---	1	Ni (1) O (4) W (1)	306.531
Ni (s) Nickel; Nickel, fcc				
0	7440-02-0	148	Ni (1)	58.6930
NTA-3 Nitrilotriacetate ion; N,N-bis(carboxymethyl)glycinate ion; Triglycollamate ion; Triglycinate ion; NTA ion				
-3	28528-44-1	751	C (6) H (6) N (1) O (6)	188.117
O2 (g) Oxygen gas; Oxygen				
0	7782-44-7	2559	O (2)	31.9988
OH-1 Hydroxide ion				
-1	14280-30-9	6558	H (1) O (1)	17.0073
OH-1 (10)_WO4-2 (6)				
-22	---	1	H (10) O (34) W (6)	1657.10
OH-1 (18)_WO4-2 (12)				
-42	---	1	H (18) O (66) W (12)	3280.18
OH-1 (2)_WO4-2				
-4	---	1	H (2) O (6) W (1)	281.852
OH-1 (7)_WO4-2 (6)				

-19	---	1	H (7) O (31) W (6)	1606.08
Pb+2 Lead(II) ion				
2	14280-50-3	2437	Pb (1)	207.200
Pb+2_WO4-2_ (s) Lead tungstate(VI)				
0	7759-01-5	2	O (4) Pb (1) W (1)	455.038
Pb (s) Lead; Lead, cubic				
0	7439-92-1	127	Pb (1)	207.200
S (s) Sulphur, alpha; Sulphur, orthorhombic; Sulphur; Sulfur, alpha; Sulfur, orthorhombic; Sulfur; Sulfur-Rhmb				
0	7704-34-9	619	S (1)	32.0600
Sr+2_WO4-2_ (s) Strontium tungstate				
0	---	1	O (4) Sr (1) W (1)	335.458
Sr (s) Strontium; Strontium, cubic; Strontium, alpha				
0	7440-24-6	55	Sr (1)	87.6200
Succinic-2 Succinate ion				
-2	56-14-4	635	C (4) H (4) O (4)	116.073
Sulfox-2 8-Hydroxy-5-quinoline sulphonate ion; 8-Hydroxy-5-quinoline sulfonate ion; 8-Hydroxyquinoline-5-sulfonate ion; 8-Hydroxyquinoline-5-sulphonate ion				
-2	40747-73-7	190	C (9) H (5) N (1) O (4) S (1)	223.203
W+3 Tungsten(III) ion; Wolfram(III) ion				
3	22541-99-7	1	W (1)	183.840
W+4_CN-1 (4)_OH-1 (4)				

-4	---	1	C(4)H(4)N(4)O(4)W(1)	355.940
W+4_CN-1(8)				
-4	---	1	C(8)N(8)W(1)	391.982
W+5_CN-1(4)_OH-1(4)				
-3	---	1	C(4)H(4)N(4)O(4)W(1)	355.940
W+5_CN-1(8)				
-3	---	1	C(8)N(8)W(1)	391.982
W(CO)6(s) Tungsten hexacarbonyl				
0	---	1	C(6)O(6)W(1)	351.902
W(s) Tungsten; Tungsten, bcc				
0	7440-33-7	42	W(1)	183.840
W2C(s) Ditungsten carbide				
0	---	1	C(1)W(2)	379.691
W2C19-3				
-3	---	1	Cl(9)W(2)	686.757
W2O5(s) Tungsten(V) oxide				
0	12036-84-9	4	O(5)W(2)	447.677
W6O21-6				
-6	---	3	O(21)W(6)	1439.03
WBr5(s) Tungsten pentabromide				
0	---	1	Br(5)W(1)	583.360
WBr6(s) Tungsten hexabromide				

0	---	1	Br (6)W(1)	663.264
WC (s) Tungsten carbide				
0	---	1	C (1)W(1)	195.851
WC12 (s) Tungsten dichloride				
0	---	1	Cl (2)W(1)	254.746
WC14 (s) Tungsten tetrachloride				
0	---	1	Cl (4)W(1)	325.652
WC15 (s) Tungsten pentachloride				
0	---	1	Cl (5)W(1)	361.105
WC16 (s) Tungsten hexachloride				
0	---	1	Cl (6)W(1)	396.558
WO+3_Cl-1 (5)				
-2	---	2	Cl (5)O (1)W(1)	377.104
WO2.72 (s) Tungsten 2.72-oxide				
0	---	1	O (2.72)W(1)	227.358
WO2.90 (s) Tungsten 2.90-oxide				
0	---	1	O (2.9)W(1)	230.238
WO2.96 (s) Tungsten 2.96-oxide				
0	---	1	O (2.96)W(1)	231.198
WO2+2				
2	---	18	O (2)W(1)	215.839

WO2+2_35DiNitrCat-2 (2)				
-2	---	1	C(12)H(4)N(4)O(14)W(1)	612.022
WO2+2_Asp-2				
0	---	2	C(4)H(5)N(1)O(6)W(1)	346.927
WO2+2_Asp-2 (2)				
-2	---	2	C(8)H(10)N(2)O(10)W(1)	478.015
WO2+2_Asp-2 (3)				
-4	---	1	C(12)H(15)N(3)O(14)W(1)	609.103
WO2+2_BenzHydroxam-1 (2)				
0	---	1	C(14)H(12)N(2)O(6)W(1)	488.099
WO2+2_Cat-2 (2)				
-2	---	1	C(12)H(8)O(6)W(1)	432.032
WO2+2_Cl-1 (3)				
-1	---	1	Cl(3)O(2)W(1)	322.198
WO2+2_Glu-2				
0	---	2	C(5)H(7)N(1)O(6)W(1)	360.954
WO2+2_Glu-2 (2)				
-2	---	2	C(10)H(14)N(2)O(10)W(1)	506.069
WO2+2_Glu-2 (3)				
-4	---	1	C(15)H(21)N(3)O(14)W(1)	651.184
WO2+2_H+1 (-2)_2OH2MePropan-1 (2)				
-2	---	1	C(8)H(12)O(8)W(1)	420.019
WO2+2_H+1 (-2)_2OHButan-1 (2)				
-2	---	1	C(8)H(12)O(8)W(1)	420.019

WO₂+2_H+1(-2)_Glycolic-1(2)				
-2	---	1	C(4)H(4)O(8)W(1)	363.911
WO₂+2_H+1(-2)_Lactic-1(2)				
-2	---	1	C(6)H(8)O(8)W(1)	391.965
WO₂+2_H+1(-2)_Mandelic-1(2)				
-2	---	1	C(16)H(12)O(8)W(1)	516.106
WO₂+2_H+1(2)_Oxalic-2(2)				
0	---	1	C(4)H(2)O(10)W(1)	393.894
WO₂+2_Malonic-2				
0	---	1	C(3)H(2)O(6)W(1)	317.885
WO₂+2_Succinic-2				
0	---	1	C(4)H(4)O(6)W(1)	331.912
WO₂(s) Tungsten(IV) oxide; Tungsten dioxide, monoclinic; Tungsten dioxide				
0	12036-22-5	5	O(2)W(1)	215.839
WO₂Cl₂(s) Tungsten dichloride dioxide				
0	---	1	Cl(2)O(2)W(1)	286.745
WO₃				
0	---	18	O(3)W(1)	231.838
WO₃_Cys-2				
-2	---	1	C(3)H(5)N(1)O(5)S(1)W(1)	350.976
WO₃_EDTA-4				
-4	---	3	C(10)H(12)N(2)O(11)W(1)	520.052
WO₃_NTA-3				
-3	---	2	C(6)H(6)N(1)O(9)W(1)	419.955

WO3_Sulfox-2				
-2	---	1	C(9)H(5)N(1)O(7)S(1)W(1)	455.041
WO3(2)_EDTA-4				
-4	---	1	C(10)H(12)N(2)O(14)W(2)	751.890
WO3(s) Tungsten(VI) oxide; Tungsten trioxide				
0	1314-35-8	5	O(3)W(1)	231.838
WO4-2 Wolframate ion; Tungstate ion				
-2	---	65	O(4)W(1)	247.838
WO4-2_EDTA-4				
-6	---	1	C(10)H(12)N(2)O(12)W(1)	536.052
WO4-2_EDTA-4(2)				
-10	---	1	C(20)H(24)N(4)O(20)W(1)	824.265
WO4-2_NTA-3				
-5	---	1	C(6)H(6)N(1)O(10)W(1)	435.954
WOC14(s) Tungsten tetrachloride oxide				
0	---	1	Cl(4)O(1)W(1)	341.651
WOF4(s) Tungsten tetrafluoride oxide				
0	---	1	F(4)O(1)W(1)	275.833
WS2(s) Tungsten disulfide; Tungstenite				
0	---	1	S(2)W(1)	247.960
Zn+2 Zinc(II) ion				
2	23713-49-7	4977	Zn(1)	65.3820

Zn+2_WO4-2_(s) Zinc tungstate				
0	---	2	O(4)W(1)Zn(1)	313.220
Zn(s) Zinc; Zinc, hexagonal				
0	7440-66-6	131	Zn(1)	65.3820